

EC-467 Mobile Application Development



BS (CE)-2021 LAB REPORT #3

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

Dr. Muhammad Bilal

LAB REPORT #3

Activity with GUI and Layout

Activity with GUI and Layout:

Creating an activity with a graphical user interface (GUI) and implementing a layout is a fundamental aspect of software development, especially in the realm of mobile applications. Whether you're developing for Android, iOS, or another platform, designing an intuitive and visually appealing interface is crucial for providing a seamless user experience. The following steps:

- 1.Setting up a new Android project in Android Studio.
- 2.Creating a new activity with a GUI layout using XML.
- 3.Implementing a basic layout structure with various UI components such as TextViews, Buttons, EditTexts, etc.
- 4.Previewing the layout within Android Studio's design editor.
- 5.Running the application on an emulator or physical device to test the GUI functionality.

Class Task:

Make a tasbeeh app to implement the text View,reset,count,view history.

Java CODE:

```
package com.huzaifashahbaz.mytasbeehapp1;

import android.os.Bundle;
import android.os.PersistableBundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    TextView tvCount;
    Button btnCount,btnReset,btnViewHistory;
```

```

int count=0;
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate( savedInstanceState );
    setContentView( R.layout.activity_main );
    tvCount=findViewById(R.id.textView);
}
public void onPressResetBtn(View View){
    count=0;
    tvCount.setText("Count:"+count);
}
public void onPressHistorybtn(View view){
    Toast.makeText( this, "JAZAKALLAJKHAIR", Toast.LENGTH_SHORT
).show();
}
public void onPressCountBtn(View View){
    count++;
    tvCount.setText("Count:"+count);
}
}

```

XML CODE:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/textView"
        android:layout_width="131dp"
        </ android:layout_height="wrap_content"
        android:text="TextView"
        android:textColorLink="#805F5F"
        tools:layout_editor_absoluteX="267dp"
        tools:layout_editor_absoluteY="142dp" />
    <LinearLayout

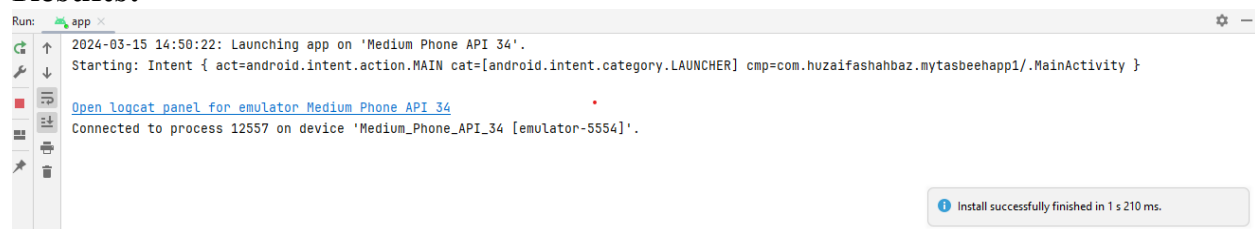
```

```

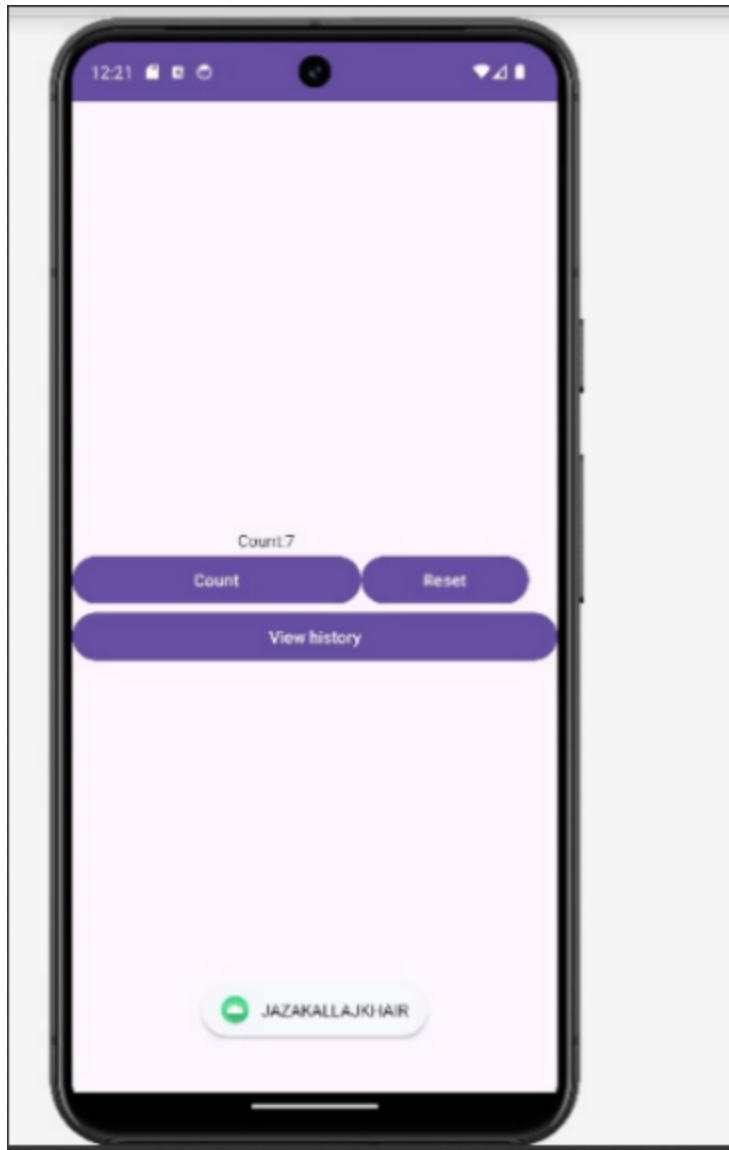
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button
            android:id="@+id/btn_count_1"
            android:layout_width="245dp"
            android:layout_height="wrap_content"
            android:onClick="onPressCountBtn"
            android:text="Count"
            tools:layout_editor_absoluteX="198dp"
            tools:layout_editor_absoluteY="156dp" />
        <Button
            android:id="@+id/btn_Rest_1"
            android:layout_width="142dp"
            android:layout_height="wrap_content"
            android:onClick="onPressResetBtn"
            android:text="Reset"
            tools:layout_editor_absoluteX="247dp"
            tools:layout_editor_absoluteY="224dp" />
    </LinearLayout>
    <Button
        android:id="@+id/btnViewhistory"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="onPressHistorybtn"
        android:text="View history"
        tools:layout_editor_absoluteX="296dp"
        tools:layout_editor_absoluteY="153dp" />
</LinearLayout>

```

Results:



Output:



HOME TASK:

Design the following.

Username

Email

Password

☒ Male ☐ Female

Register

Already Registered?
Login Here

XML CODE:

<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

android:layout_height="match_parent"

android:orientation="vertical"

android:padding="16dp">

<EditText

android:id="@+id/usernameEditText"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:hint="Username"

android:minHeight="48dp" />

<EditText

android:id="@+id/emailEditText"

```
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Email"
android:inputType="textEmailAddress"
android:minHeight="48dp" />
```

```
<EditText
    android:id="@+id/passwordEditText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:minHeight="48dp" />
```

```
<RadioGroup
    android:id="@+id/genderRadioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:onClick="onPressRadioGroupbtn">
```

```
<RadioButton
    android:id="@+id/maleRadioButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="onPressMalebtn"
    android:text="Male" />
```

```
<RadioButton
    android:id="@+id/femaleRadioButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="onPressFemalebtn"
    android:text="Female" />
```

```
</RadioGroup>
```

```
<Button
    android:id="@+id/registerButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```
    android:layout_gravity="center_horizontal"
    android:onClick="onPressRegisterbtn"
    android:text="Register" />
```

```
<TextView
    android:id="@+id/alreadyRegisteredTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center_horizontal"
    android:text="Already Registered?" />
```

```
<Button
    android:id="@+id/loginButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center_horizontal"
    android:onClick="onPressLoginHerebtn"
    android:text="Login Here" />
```

```
</LinearLayout>
```

Java CODE:

```
package com.huzaifashahbaz.madlab3;
public class RegistrationActivity extends AppCompatActivity {
    private EditText mUsernameEditText;
    private EditText mEmailEditText;
    private EditText mPasswordEditText;
    private RadioGroup mGenderRadioGroup;
    private Button mRegisterButton;
    private TextView mAlreadyRegisteredTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mUsernameEditText = findViewById(R.id.usernameEditText);
        mEmailEditText = findViewById(R.id.emailEditText);
        mPasswordEditText = findViewById(R.id.passwordEditText);
        mGenderRadioGroup = findViewById(R.id.genderRadioGroup);
        mRegisterButton = findViewById(R.id.registerButton);
        mAlreadyRegisteredTextView =
        findViewById(R.id.alreadyRegisteredTextView);
    }
}
```

```

mRegisterButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        String username = mUsernameEditText.getText().toString();
        String email = mEmailEditText.getText().toString();
        String password = mPasswordEditText.getText().toString();
        int genderId = mGenderRadioGroup.getCheckedRadioButtonId();
        RadioButton genderRadioButton = findViewById(genderId);
        String gender = genderRadioButton.getText().toString();

        // Validate form input
        if (username.isEmpty() || email.isEmpty() || password.isEmpty() ||
gender.isEmpty()) {
            Toast.makeText(RegistrationActivity.this, "Please fill out all fields",
Toast.LENGTH_SHORT).show();
            return;
        }

        // TODO: Send registration data to server or database

        // Clear form input
        mUsernameEditText.setText("");
        mEmailEditText.setText("");
        mPasswordEditText.setText("");
        mGenderRadioGroup.clearCheck();

        Toast.makeText(RegistrationActivity.this, "Registration successful!",
Toast.LENGTH_SHORT).show();
    }
});

mAlreadyRegisteredTextView.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View view) {
        // TODO: Navigate to login screen
    }
});
}

```

```

}
public class RegistrationActivity extends AppCompatActivity {
    private EditText mUsernameEditText;
    private EditText mEmailEditText;
    private EditText mPasswordEditText;
    private RadioGroup mGenderRadioGroup;
    private Button mRegisterButton;
    private TextView mAlreadyRegisteredTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_registration);

        mUsernameEditText = findViewById(R.id.usernameEditText);
        mEmailEditText = findViewById(R.id.emailEditText);
        mPasswordEditText = findViewById(R.id.passwordEditText);
        mGenderRadioGroup = findViewById(R.id.genderRadioGroup);
        mRegisterButton = findViewById(R.id.registerButton);
        mAlreadyRegisteredTextView =
        findViewById(R.id.alreadyRegisteredTextView);

        mRegisterButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String username = mUsernameEditText.getText().toString();
                String email = mEmailEditText.getText().toString();
                String password = mPasswordEditText.getText().toString();
                int genderId = mGenderRadioGroup.getCheckedRadioButtonId();
                RadioButton genderRadioButton = findViewById(genderId);
                String gender = genderRadioButton.getText().toString();

                // Validate form input
                if (username.isEmpty() || email.isEmpty() || password.isEmpty() ||
gender.isEmpty()) {
                    Toast.makeText(RegistrationActivity.this, "Please fill out all fields",
Toast.LENGTH_SHORT).show();
                    return;
                }
            }
        });
    }
}

```

```

        // TODO: Send registration data to server or database

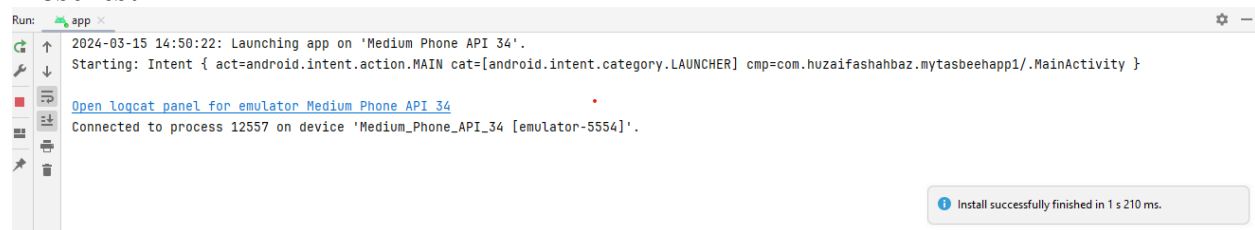
        // Clear form input
        mUsernameEditText.setText("");
        mEmailEditText.setText("");
        mPasswordEditText.setText("");
        mGenderRadioGroup.clearCheck();

        Toast.makeText(RegistrationActivity.this, "Registration successful!",
        Toast.LENGTH_SHORT).show();
    }
});

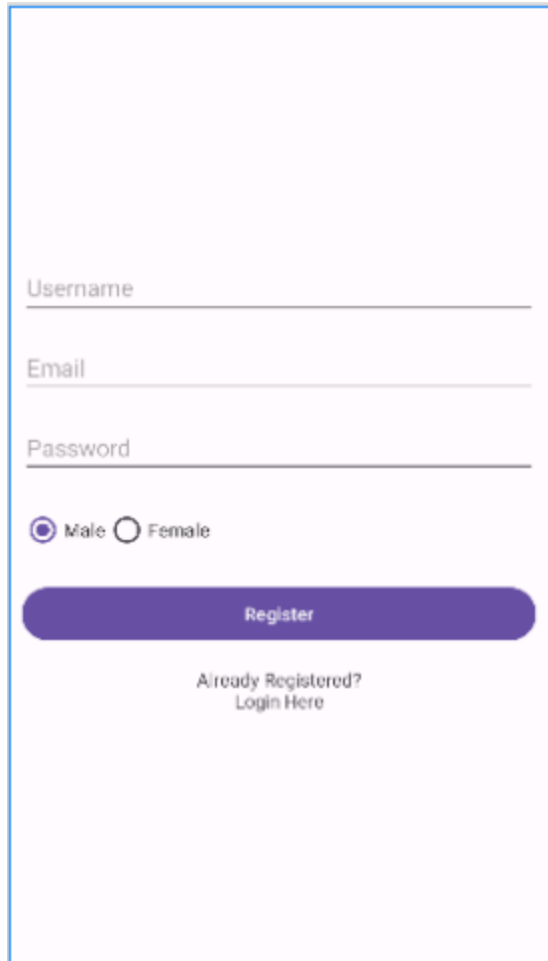
mAlreadyRegisteredTextView.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View view) {
        // TODO: Navigate to login screen
    }
});
}
}
}

```

Results:



OUTPUT:

A registration form with a light pink background and a blue border. It contains three input fields for 'Username', 'Email', and 'Password'. Below these is a gender selection section with 'Male' (selected) and 'Female' radio buttons. A blue 'Register' button is centered below the gender options. At the bottom, the text 'Already Registered? Login Here' is displayed in a smaller font.

Username

Email

Password

☒ Male ☐ Female

Register

Already Registered?
Login Here

Conclusion:

In conclusion, we had know how to make the tasbeeh counter app and registration form by performing java code and xml code(Linear Layout).Also we saw their working ,construction and functions in frontend and backend of the mobile app.